

Md Motaleb Hossen Manik

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Research Statement

I am currently a **Ph.D. student in Computer Science** at **Rensselaer Polytechnic Institute**, where my research focuses on the **evaluation and reliability of AI systems**, particularly large language models, vision-language models, and agentic AI. I study how AI outputs and actions can be **measured, structured, verified, and audited under realistic conditions**, including deployment-aware evaluation, structured reliability reporting, semantic consistency, provenance, and safe agent behavior. I investigate these questions across medical and educational AI, scientific knowledge workflows, and autonomous-agent systems. I am seeking a **tenure-track faculty position** or **research scientist role** to advance reliable, verifiable, and accountable AI systems.

Academic Credentials

Ph.D. in Computer Science

Aug 2024 – Present

Rensselaer Polytechnic Institute (RPI), Troy, NY 12180, USA

Advisor: Prof. Ge Wang, Department of Biomedical Engineering, with affiliation in Computer Science, RPI

Research Group: Wang-AXIS Lab, RPI

M.Sc. Eng. in Computer Science and Engineering

Sep 2020 – Mar 2023

Khulna University of Engineering & Technology (KUET), Khulna, Bangladesh

CGPA: 4.00/4.00

Advisor: Prof. Dr. Kazi Md. Rokibul Alam, Department of Computer Science and Engineering, KUET

Thesis: *A Study on a Blockchain-Based Secure Framework for Multi-party Skyline Queries*

B.Sc. Eng. in Computer Science and Engineering

Nov 2015 – Mar 2020

Khulna University of Engineering & Technology (KUET), Khulna, Bangladesh

CGPA: 3.93/4.00

Rank: 1st out of 134 graduates

Advisor: Prof. Dr. M. M. A. Hashem, Department of Computer Science and Engineering, KUET

Thesis: *A Study on Sentiment Analysis of Bangla and Phonetic Bangla Reviews*

Teaching Experience

Teaching Assistant

Aug 2024 – Spring 2026

Department of Computer Science, Rensselaer Polytechnic Institute (RPI), Troy, NY

Courses:

- CSCI 2600 *Principles of Software* (Spring 2025, Summer 2025, Spring 2026)
- CSCI 4430 *Programming Languages* (Fall 2025)
- CSCI 4440 *Software Design and Documentation* (Summer 2025)
- CSCI 1700/2700/4968/4970 *RCOS* (Fall 2024, Spring 2025), including evaluation of student open-source project presentations

Assistant Professor (On Study Leave)

Jun 2024 – Present

Department of Computer Science and Engineering, KUET, Bangladesh

Department of Computer Science and Engineering, KUET, Bangladesh

Courses:

- CSE 2103/2104 *Microprocessors and Microcontrollers* (theory and lab)
- CSE 3206 *Robotics Laboratory* (lab)
- CSE 3217 *Mobile Computing* (theory)
- CSE 4120 *Technical Writing and Seminar* (lab)
- CSE 4223/4224 *Digital System Design* (theory and lab)

Research Experience

Evaluation, Reliability & Deployment of Language and Vision–Language Models 2024 – Present

Ph.D. research, Rensselaer Polytechnic Institute

I develop methods for **measuring AI reliability beyond task accuracy**. In **Unified Deployment-Aware Evaluation of Open Reasoning Language Models** (TMLR, in press 2026), I evaluated seven model configurations across four benchmarks and three prompting strategies (**84 conditions; 19,992 examples**), jointly analyzing performance, uncertainty, latency, memory, prompt sensitivity, and deployment trade-offs. I also developed **HypothesisMed** for structured reliability reporting, **MILU** for multimodal semantic evaluation, and earlier comparative work on **ChatGPT vs. DeepSeek** for AI-based code generation.

Provenance, Auditability & Secure Knowledge Systems 2020 – Present

M.Sc. foundation at KUET; continuing Ph.D. research at RPI

I develop **provenance and integrity mechanisms** that make digital and AI-generated knowledge traceable, tamper-evident, reproducible, and securely exchanged. **SlideChain** provides blockchain-backed semantic provenance for structured VLM outputs over **1,117** lecture slides, with **100% tamper detection** and deterministic reproducibility under controlled testing. Earlier work includes my M.Sc. research on secure multi-party skyline queries and **ShaEr**, a privacy-preserving medical-data sharing framework.

Safe Agentic AI & Scientific Knowledge Governance 2025 – Present

Ph.D. research, Rensselaer Polytechnic Institute

I study how **autonomous AI systems behave and act under permissions, policies, social interaction, and governance constraints**. My workflow-agent research evaluates permission scope, semantic policy gating, and tamper-evident execution records; **OpenClaw** studies risky instruction propagation and norm enforcement across **14,490 autonomous agents**; and **ADAPT** investigates adaptive governance in AI-mediated scholarly publishing, leading to **U.S. Provisional Patent Application No. 63/975,609**.

Multimodal AI for Education & Medical Workflows 2025 – Present

Ph.D. research and system development, Rensselaer Polytechnic Institute

I develop **multimodal and interactive AI systems for education and medical knowledge workflows**. I helped build **MEDI-SLATE**, comprising **1,117 slides** and **262,182 narration tokens** from a 23-lecture medical imaging course; contributed to **ALIVE**, a fully local interactive lecture system; and co-developed an **optically emulated CT (OECT) scanner** for hands-on medical imaging education.

Foundations in Machine Learning & Natural Language Processing 2019 – 2025

Earlier research, Khulna University of Engineering & Technology and RPI

My earlier work established a foundation in **machine learning, NLP, and robustness**, including Bangla and phonetic-Bangla sentiment analysis, unified sentiment and emotion recognition, adversarially robust text classification, and closed-domain question answering. Related methodological work includes **N-ReLU**, a stochastic extension of the ReLU activation function.

Journal Articles

2026

- **Manik, M. M. H.**, and Wang, G. “Unified Deployment-Aware Evaluation of Open Reasoning Language Models.” *Transactions on Machine Learning Research (TMLR)*, 2026. Link.
- Habib, M. A., and **Manik, M. M. H.** “Blockchain-Enabled Secure Land Record Management System with a New Lightweight Cryptosystem Based on a Hybrid Chaotic Map.” *Blockchain: Research and Applications*, 100531, 2026. Link.
- **Manik, M. M. H.**, and Wang, G. “Evaluating Risk and Auditability in Workflow Agents for Safety-Critical Domains.” *Journal of Information and Intelligence*, in press, 2026. Link.
- **Manik, M. M. H.**, Islam, M. Z., and Wang, G. “MEDI-SLATE: Medical Imaging Slide–Lecture Aligned Teaching Ensemble.” *Visual Computing for Industry, Biomedicine, and Art*, 9:8, 2026. Link.
- **Manik, M. M. H.**, Islam, M. Z., and Wang, G. “MILU: A Consensus Ensemble Benchmark for Multimodal Medical Imaging Lecture Understanding.” *Journal of Medical Imaging*, 13(6):062202, 2026. Link.
- **Manik, M. M. H.**, Muldowney, W., Islam, M. Z., and Wang, G. “Development of an Optically Emulated Computed Tomography Scanner for College Education.” *Visual Computing for Industry, Biomedicine, and Art*, 9:2, 2026. Link.

2025

- Ahmed, M. R., Aziz, A., **Manik, M. M. H.**, and Habib, M. A. “Blockchain-aided Comparative Study of Heart Disease Detection Using Machine Learning-Based Approaches with an Expanded Dataset.” *Computers in Biology and Medicine*, 2025. Link.
- Habib, M. A., and **Manik, M. M. H.** “ShaEr: A Blockchain-Based Framework for Secure Medical Data Sharing and Monetisation.” *IET Blockchain*, 2025. Link.

2023

- **Manik, M. M. H.** “A Novel Approach in Determining Areas to Lockdown During a Pandemic: COVID-19 as a Case Study.” *International Journal of Information Engineering and Electronic Business*, 15(2):30, 2023. Link.

2021

- Rahman, M. M., Islam, M., **Manik, M. M. H.**, Islam, M. R., and Al-Rakhami, M. S. “Machine Learning Approaches for Tackling Novel Coronavirus (COVID-19) Pandemic.” *SN Computer Science*, 2:1–10, 2021. Link.

Conference Papers

2026

- **Manik, M. M. H.** and Wang, G. “NormBench: A Two-Stage Benchmark for Measuring Compliance and Regulatory Behavior in Local LLM Societies.” *38th International Conference on Tools with Artificial Intelligence (ICTAI)*, 2026. Accepted.

2025

- **Manik, M. M. H.**, Sagor, A. H., Mondal, F. A., Touhid, M. M., and Islam, M. Z. “Unifying Sentiment Analysis and Emotion Recognition for Bangla Text: A Hybrid Approach.” *International Conference on Electrical, Computer and Communication Engineering (ECCE)*, 2025. Link.

2024

- Atik, A. I., Rahman, A., Sami, S. A., and **Manik, M. M. H.** “An Android Application for Healthcare Management System with ML-Driven Solution.” *27th International Conference on Computer and*

Information Technology (ICCIT), 2024. Link.

- Saha, A., Majumder, S., **Manik, M. M. H.**, and Hashem, M. A. “Enhancing Hyperledger Fabric: A Scalable Framework for Optimized Blockchain Performance.” *27th International Conference on Computer and Information Technology (ICCIT)*, 2024. Link.
- Islam, M. R., Alam, K. M. R., and **Manik, M. M. H.** “Decentralized GDPR Compliance: A Blockchain Framework for Personal Data Management.” *27th International Conference on Computer and Information Technology (ICCIT)*, 2024. Link.

2023

- **Manik, M. M. H.**, Mahadi, J., Touhid, M. M., and Alam, K. M. R. “Enhancing Robustness of Machine Learning Algorithms for Bangla Text Classification: A Defensive Approach against Adversarial Attacks.” *6th International Conference on Electrical Information and Communication Technology (EICT)*, 2023. Link.
- **Manik, M. M. H.**, Sagor, A. H., Habib, M. A., Touhid, M. M., Ahmed, T., Islam, M. Z., and Haque, F. “Redefining Crime Record Storage: An Advanced Architecture Harnessing the Power of Blockchain Technology.” *26th International Conference on Computer and Information Technology (ICCIT)*, 2023. Link.
- Majumder, S., Zaha, R., **Manik, M. M. H.**, and Alam, K. M. R. “A Blockchain Based Scalable Framework for Academic Document Verification.” *26th International Conference on Computer and Information Technology (ICCIT)*, 2023. Link.
- Habib, M. A., **Manik, M. M. H.**, and Zaman, S. “A Blockchain-Based Technique to Prevent Grade Tampering: A University Perspective.” *International Conference on Electrical, Computer and Communication Engineering (ECCE)*, 2023. Link.
- Habib, M. A., and **Manik, M. M. H.** “A Technique to Avoid Blockchain Denial of Service (BDoS) and Selfish Mining Attack.” *Fifth International Conference on Blockchain Computing and Applications (BCCA)*, 2023. Link.

2022

- **Manik, M. M. H.**, Haque, F., Hashem, M. M. A., Habib, M. A., Islam, M. Z., and Ahmed, T. “A Hybrid Framework for Sentiment Analysis from Bangla Texts.” *25th International Conference on Computer and Information Technology (ICCIT)*, 2022. Link.
- **Manik, M. M. H.**, Alam, K. M. R., and Morimoto, Y. “A Blockchain Based Secure Framework for User-Centric Multi-Party Skyline Queries.” *25th International Conference on Computer and Information Technology (ICCIT)*, 2022. Link.
- **Manik, M. M. H.**, Habib, M. A., and Ahmed, T. “Machine Learning Algorithms on COVID-19 Prediction Using CpG Island and AT-CG Feature on Human Genomic Data.” *International Conference on Machine Intelligence and Emerging Technologies*, 2022. Link.

2020

- Rahman, M. M., **Manik, M. M. H.**, Islam, M. M., Mahmud, S., and Kim, J. H. “An Automated System to Limit COVID-19 Using Facial Mask Detection in Smart City Network.” *IEEE International IOT, Electronics and Mechatronics Conference (IEMTRONICS)*, 2020. Link.

2019

- Haque, F., **Manik, M. M. H.**, and Hashem, M. M. A. “Opinion Mining from Bangla and Phonetic Bangla Reviews Using Vectorization Methods.” *4th International Conference on Electrical Information and Communication Technology (EICT)*, 2019. Link.

Preprints

2026

- **Manik, M. M. H.**, and Wang, G. “Toward AI-Empowered Higher Education: A Workflow-Oriented Review and Perspective.” *EdArXiv*, 2026. Link.

- **Manik, M. M. H.**, and Wang, G. “HypothesisMed: Inference-Time Answer Fusion and Structured Hypothesis-Space Reporting for Biomedical Question Answering.” arXiv:2606.00971, 2026. Link.
- **Manik, M. M. H.**, and Wang, G. “OpenClaw Agents on Moltbook: Risky Instruction Sharing and Norm Enforcement in an Agent-Only Social Network.” arXiv:2602.02625, 2026. Link.

2025

- **Manik, M. M. H.**, Islam, M. Z., and Wang, G. “SlideChain: Semantic Provenance for Lecture Understanding via Blockchain Registration.” arXiv:2512.21684, 2025. Link.
- Islam, M. Z., **Manik, M. M. H.**, and Wang, G. “ALIVE: An Avatar-Lecture Interactive Video Engine with Content-Aware Retrieval for Real-Time Interaction.” arXiv:2512.20858, 2025. Link.
- **Manik, M. M. H.**, Islam, M. Z., and Wang, G. “N-ReLU: Zero-Mean Stochastic Extension of ReLU.” arXiv:2511.07559, 2025. Link.
- **Manik, M. M. H.** “ChatGPT vs. DeepSeek: A Comparative Study on AI-Based Code Generation.” arXiv:2502.18467, 2025. Link.

2024

- **Manik, M. M. H.**, Habib, M. A., Islam, M. Z., Ahmed, T., and Haque, F. “Analyzing the Dynamics of COVID-19 Lockdown Success: Insights from Regional Data and Public Health Measures.” arXiv:2402.18594, 2024. Link.
- Roy, S. C., and **Manik, M. M. H.** “Question-Answering System for Bangla: Fine-Tuning BERT-Bangla for a Closed Domain.” arXiv:2410.03923, 2024. Link.

2022

- Habib, M. A., and **Manik, M. M. H.** “Classification of DNA Sequence Using Machine Learning Techniques.” EasyChair preprint, 2022. Link.

Submitted Manuscripts

- **Manik, M. M. H.**, Alam, K. M. R., and Morimoto, Y. “A Secure Framework for User-Centric Multiparty Skyline Queries via Pruned and Prioritized Datasets.” Submitted to *Blockchain Research and Applications*; under review.
- **Manik, M. M. H.**, and Wang, G. “ADAPT: AI-Driven Decentralized Adaptive Publishing Transformer.” Submitted to *Research Integrity and Peer Review*; under review.
- **Manik, M. M. H.**, and Wang, G. “Emergent Decentralized Regulation in a Purely Synthetic Society.” Submitted to *Scientific Reports*; under review. Preprint.

Honors & Awards

- **Best Paper Award**, International Conference on Electrical, Computer and Communication Engineering (ECCE), IEEE, 2023.
- **Dean’s Award** (three consecutive years), Faculty of Electrical and Electronic Engineering, KUET, awarded for top academic performance among all CSE students.
- **Merit Scholarship**, KUET (2016–2020), granted to students ranking in the top 5% of their cohort.

Grants & Funding

- **AI-Integrated Curriculum Development Award** (2026). Funded project: *AI-Integrated Interactive Classroom Platform (BMED 2300)*. Award amount: **\$5,000**. Funding period: **May 15–August 15, 2026**. PI: Prof. Ge Wang (RPI); Role: Graduate Research Collaborator.
- **NVIDIA Academic Grant Program** (2025). Hardware grant: $1 \times$ NVIDIA DGX Spark. Project: *Iterative Avatar Teaching in the NVIDIA Omniverse*. PI: Prof. Ge Wang (RPI); Role: Graduate Research Collaborator.

Patents

- **Manik, M. M. H.**, and Wang, G. “ADAPT: AI-Driven Decentralized Adaptive Publishing Transformer.” U.S. Provisional Patent Application No. 63/975,609, filed February 4, 2026. Rensselaer Polytechnic Institute.

Professional Service

Peer Reviewer

Journals

- **IEEE Transactions on Computational Imaging**, IEEE, 2026.
- **Engineering Reports**, Wiley, 2026.
- **Technology in Society**, Elsevier, 2025.
- **Journal of Systems and Software**, Elsevier, 2025.

Conferences

- IEEE COMPAS 2025; EICT 2025; ICCAconf 2024; EICT 2023.

Mentorship & Student Supervision

- Supervised 3 undergraduate B.Sc. thesis groups and 2 undergraduate project groups at the Department of Computer Science and Engineering, KUET (2022–2024), covering topics in machine learning, blockchain systems, and large language models.

Invited Talks & Presentations

- “**All About AI: Avatarized Online Course**,” ECSE Best Practices Series, Department of Electrical, Computer and Systems Engineering, Rensselaer Polytechnic Institute, August 7, 2025. Invited by Prof. Shayla Sawyer (Director, Mercer XLab, RPI). Presented the ALIVE avatarized lecture system to a cross-departmental faculty audience and received commendation from senior RPI faculty for its significance and experimental potential.

Technical Skills

Machine Learning, LLMs & Multimodal AI

Large language models, vision language models, multimodal learning, transformer architectures, retrieval augmented generation, prompt engineering, benchmark design, model evaluation, scientific slide understanding, medical imaging AI.

Frameworks & Libraries

PyTorch, TensorFlow, HuggingFace Transformers, SentenceTransformers, Scikit-learn, NumPy, Pandas, Matplotlib, FAISS, Whisper ASR, local LLM pipelines, structured JSON evaluation workflows.

Agentic Systems & AI Safety

Synthetic agent societies, workflow agents, norm enforcement, action-inducing risk modeling, auditability, decentralized governance, trustworthy AI evaluation.

Blockchain, Security & Provenance

Ethereum, Solidity, Hyperledger Fabric, Hardhat, IPFS, smart contracts, cryptographic hashing, Keccak-256 commitments, decentralized storage, blockchain provenance, secure medical data sharing, research-integrity workflows.

Programming & Tools

Python, C/C++, Java, JavaScript, SQL; Git, Linux/Bash, Jupyter Notebook, REST APIs, JSON pipelines, L^AT_EX, reproducible experiment logging.